



Adolescent Girls' Musculoskeletal Pain is More Affected by Insomnia Than Boys, and Through Different Psychological Pathways

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Abstract: Prior research has established that insomnia is predictive of pain in adolescents and that psychological mechanisms have a crucial role in this relationship. Adolescent girls report more insomnia and pain than boys, yet little is known of gender differences in how insomnia influences pain. This study assessed gender differences in levels and trajectories of insomnia and pain during adolescence, and whether rumination and negative mood mediated the effect of insomnia on pain. Longitudinal survey data measured on 5 annual occasions ($N^{\text{baseline}} = 2,767$) were analyzed in a multigroup longitudinal serial mediation model. A final model was generated with insomnia as the predictor, rumination and depressed mood as mediators, pain as the outcome, and gender as the grouping variable. The results showed that insomnia predicted pain in adolescents, with an effect 3.5 times larger in girls than boys. Depressed mood was the main mediator in boys. In girls, rumination was the only significant mediator. There were significant gender differences in the effects of insomnia on rumination and pain, and in the effects of rumination on depressed mood and pain, with stronger effects in girls. These results highlight that girls and boys should be considered separately when studying the relationship between insomnia and pain.

Perspective: Levels of insomnia and pain are progressively higher in adolescent girls than boys, across adolescence. The predictive strength of insomnia symptoms for future pain is 3.5 times greater in girls, with distinct gender-specific underlying pathways: rumination partially mediates this effect in girls, while depressed mood does so in boys.

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Insomnia is common in community samples of adolescents, with prevalences estimated at 20 to 40%.^{1,2} Its co-occurrence among those with chronic musculoskeletal pain is over 50%,^{3,4} and this comorbidity is associated with considerable pain-related disability.^{5,6} Upward of 45% of children and adolescents suffer from chronic musculoskeletal pain,^{7,8} and this prevalence has increased over time,^{8,9} partially explained by increases in sleep disturbance and negative affect.¹⁰ Adolescent chronic pain is likely to persist into adulthood¹¹ where it

is among the leading predictors of years lived with disability globally.¹² Ample evidence supports a bidirectional relationship between insomnia and pain across adolescence^{13–15} and insomnia is a stronger predictor of subsequent pain than vice-versa. In order to prevent a negative spiral of mutually increasing symptom levels, it is critical to understand how insomnia, from an early age, influences pain. Although still an emerging research field, negative mood and affect appear to be key mechanisms that mediate the effect of insomnia on pain but not the effect of pain on insomnia.^{16,17}

The response styles theory¹⁸ proposes that early increases in rumination explain subsequent increases in mood symptoms in adolescence.¹⁹ Rumination is a maladaptive problem-solving response²⁰ and a transdiagnostic factor that is embedded in the etiology of insomnia,²¹ chronic pain, depressed mood,²² and anxious mood.²³ Specifically, insomnia leads to increased rumination by impairing adaptive attentional control, which leads to sustained attention to negative thoughts and emotions.²⁴ This increase in rumination may elevate negative moods,¹⁸ which in turn are associated with more pain.¹⁶ Rumination, which is functionally analogous with pain-catastrophizing, is also a stand-alone predictor of more severe pain.²⁰ As a result, rumination may either, as the response styles theory suggests, mediate the effect of insomnia on pain through its effect on negative mood; or independently influence negative mood due to its overlap with pain-catastrophizing.

The possibility that sex or gender moderates the sleep-pain relationship has largely been neglected in previous research,^{25,26} although it has been proposed as an important factor to consider.²⁵ For example, one previous study found the effect of insomnia on pain to be stronger in adolescent girls than in boys,²⁷ while another study reported the reverse relationship.²⁸ Adolescent girls also report greater musculoskeletal pain than boys⁷ and more severe insomnia symptoms,¹ as well as more negative moods^{29,30} and maladaptive emotion regulation strategies such as rumination.¹⁹ These gender differences appear with advancing pubertal development, due to developmental changes that are unique to this developmental stage.^{25,31,32} During early adolescence, girls are twice as likely to ruminate as boys,¹⁹ and in late adolescence, girls are twice as likely to experience anxiety³⁰ and depression.²⁹ The response styles theory proposes that the greater early tendency of girls to ruminate explains the later gender differences in negative moods.¹⁹ Given that insomnia influences rumination, and negative mood influences pain, this process may explain a stronger insomnia-pain relationship in girls. It is important to explore potential gender differences in the mechanisms underlying the insomnia-pain relationship, as not accounting for gender-related differences may have confounded the results of previous research. Increased knowledge of the gender-specific role of rumination and negative mood in the insomnia-pain relationship might also inform future treatment strategies.

The first aim of this study was to map out developmental trajectories of insomnia and pain across adolescence (5 annual measurements from 13/14 to 17/18

years). The second aim was to assess the roles of rumination, depressed mood, and anxious mood as mediators in the insomnia-pain relationship. Specifically, we considered serial mediation and examined whether insomnia influenced rumination; whether that had an effect on depressed and anxious mood; and whether that had an effect on subsequent pain. Our third aim was to examine if the trajectories and effects differed by gender, hypothesizing that the symptom levels and increases over time are greater in girls than boys; that the effect of insomnia on pain is stronger in girls; and that the serial mediation of rumination and negative mood partly explains this difference.

Methods

Participants and Procedure

The current study is based on data from the Three Cities Study, a longitudinal project that used a prospective age-cohort design to broadly assess transdiagnostic factors that may explain mental ill-health in adolescents across 5 annual waves of measurements from 2014 to 2018. The study targeted all adolescents living in 3 medium-sized municipalities in mid-Sweden. As the target population was the general population of Swedish adolescents, we did not impose any exclusion criteria in this study, but rather opted to include all available participants. Although unobserved, the study sample can be assumed to include clinical subpopulations with mental or somatic disorders. The mean age was 13.65 (standard deviation [SD] = .65) years during the first (baseline) measurement wave and 17.68 (SD = .66) years during the fifth wave. The study required active consent from the adolescents and passive consent from their parents. The adolescents completed self-report questionnaires in their classrooms during school hours. The teachers were asked to leave the classroom while a trained test leader was present. If an adolescent did not wish to participate, they were informed that they could leave the room, or that they could stay but abstain from filling out the form.

The original target sample comprised 3,336 seventh- and eighth-grade students, of whom 2,767 participated in the first (baseline: T1) measurement wave. These comprise the analytic sample. Of the baseline sample, 2,521 adolescents participated in wave 2 (T2), 1,982 participated in wave 3 (T3), 1,666 participated in wave 4 (T4), and 1,568 participated in wave 5 (T5). The participation rate per measurement occasion is detailed in Fig 1. The study was approved by the regional ethical review board in Uppsala, Sweden (dnr. 2013/384). Further details on the Three Cities Study are published elsewhere.³³

Previous studies published using data from the Three Cities Study focused on a range of topics. These include parent behaviors,³⁴ self-harm,³⁵ sleep duration in relation to technology usage³⁶ and impulsivity,³⁷ and as a mediator between distress and rumination.³⁸ They also include studies of stress,^{39,40} social anxiety,⁴¹ and the

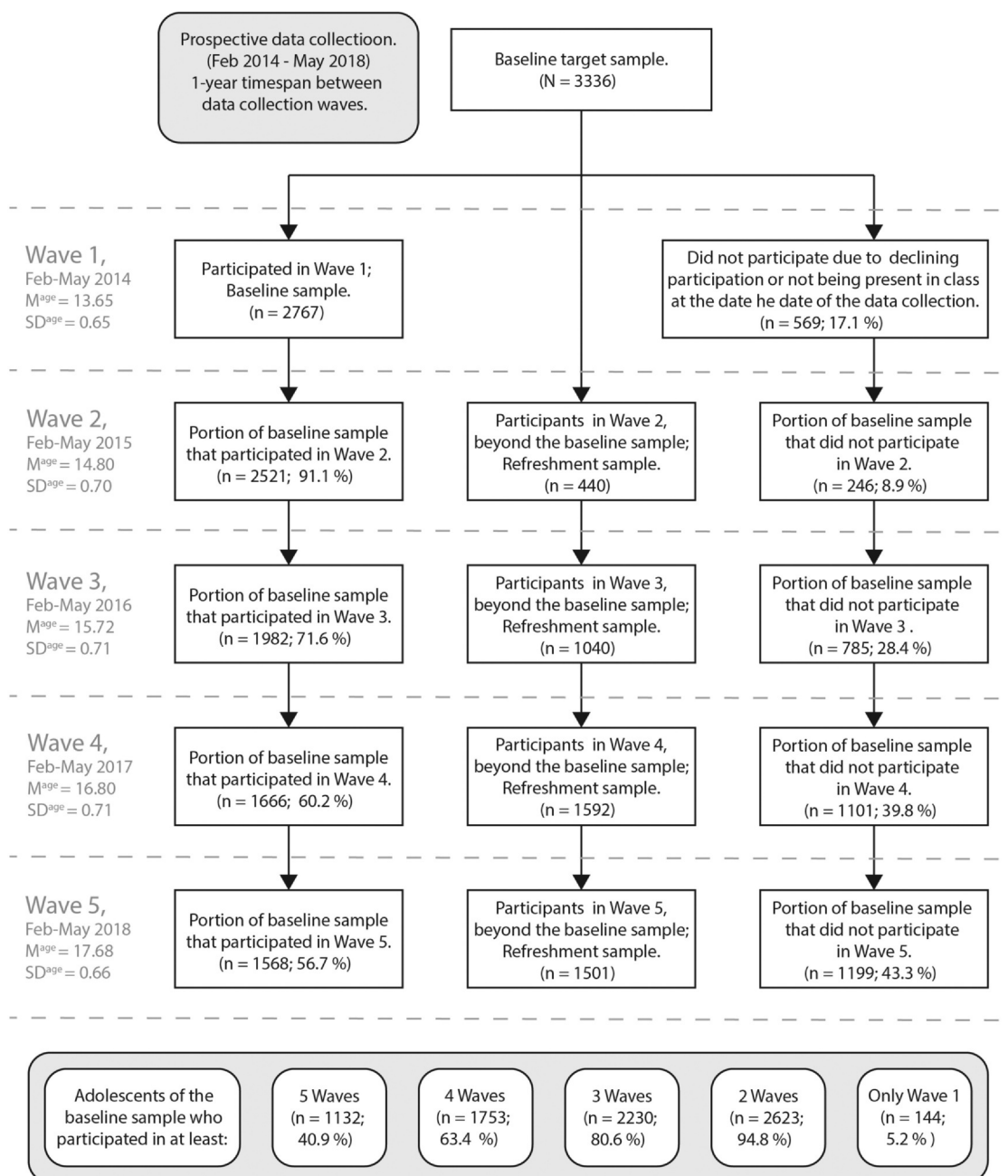


Figure 1. The study flowchart, in line with the Strengthening the Reporting of Observational Studies in Epidemiology statement (<https://www.strobe-statement.org/>). The indicators of mean age to the left of the figure refer to the baseline sample.

relationship between stress and pain.⁴² Two previous studies using this dataset concerned the relationship between insomnia and pain.^{17,43} A novelty of the current study is that it focuses on the interconnection between rumination and negative mood in their longitudinal effect of insomnia on pain, and that it explicitly examines gender differences in this effect. In addition, whereas a previous study has examined how insomnia and pain co-develop in adolescents, the

current study aimed to assess differences in symptom levels and changes between adolescent girls and boys.

An ordinal logistic regression analysis, with rate of attrition as the outcome, indicated that greater age ($b = .343$, $P < .001$), being born outside Sweden or having parents born outside Sweden ($b = .180$, $P < .001$), and greater symptoms of insomnia ($b = .046$, $P < .001$) and rumination ($b = .0001$, $P = .041$) were associated with a greater likelihood of dropout.

Measures

Sociodemographic Measures

Age was recorded as the respondent's chronological age in years. *Socioeconomic status* (SES) was determined using the Family Affluence Scale II from the World Health Organization's Health Behavior in School-aged Children survey,⁴⁴ dichotomized as either lower and equal to, or higher than, 4.61 (ie, 1 SD below the population average of Swedish adolescents).⁴⁴ When used as a covariate, the full Family Affluence Scale II score was used. *Immigrant status* was categorized into 4 categories, in line with the official Swedish definition: either 1) as being born in Sweden with at least 1 parent having been born in Sweden; 2) being born outside Sweden or having both parents born outside Sweden, but within the European Union; 3) being born, or having both parents born, outside the European Union but within Europe; or 4) being born outside of Europe or having both parents born outside of Europe. Immigrant status⁴⁵ is the closest proxy to race and ethnicity available in official Swedish census data.⁴⁶

Sleep duration was measured via the School Sleep Habits Survey for adolescents.⁴⁷ The participants were asked to indicate when they usually go to bed, how long it usually takes them to fall asleep, and when they usually wake up in the morning, on weekdays and at weekends, during the last 2 weeks. Sleep duration was calculated separately for weekdays and weekends, as the mean time from bedtime till wake time, subtracting the reported time it took to fall asleep. The cutoff for insufficient sleep duration was set at <8 hours of sleep per night, based on the recommendations of the American Academy of Sleep Medicine that adolescents aged 13 to 18 years old should sleep on average for 8 to 10 hours per night.⁴⁸

Binary gender was measured by a dichotomous item at baseline (T1), where the participants were asked to indicate whether they were a "girl" or a "boy" by checking a box next to each word on the survey's front page. As the measure was not originally designed to measure sex or gender differences, being a self-report binary measure, the distinction between sex and gender in the variable is ambiguous. However, in line with the Sex and Gender Equity in Research guidelines,⁴⁹ we believe that the importance of assessing sex and gender differences in the insomnia-pain relationship in adolescents outweighs the limitations of the measure.

Insomnia

Symptoms of insomnia were measured on the Insomnia Severity Index.⁵⁰ The measure consists of 7 5-point Likert scale items, ranging from 0 ("no problems at all") to 4 ("very much a problem"), regarding difficulties in falling or staying asleep, and also daytime distress and impairment due to sleep problems, during the last 2 weeks. The internal consistency was satisfactory in a sample of Swedish adolescents ($\alpha = .83$).⁵¹ The cutoff for clinical insomnia in adolescents is 9.⁵² In the current sample, the internal consistency was $\alpha = .85$ for

T1, $\alpha = .84$ for T2, $\alpha = .86$ for T3, $\alpha = .87$ for T4, and $\alpha = .88$ for T5.

Musculoskeletal Pain

Musculoskeletal pain was defined as "pain in the back, neck or shoulders" and measured by 3 aspects of the experience of pain:

Pain intensity was measured on a numeric rating scale, ranging from 0 ("not painful at all") to 9 ("very painful").⁵³ Respondents were asked to rate the average intensity of their pain during the last 6 months.

Pain frequency was measured by one item from the World Health Organization's Health Behavior in School-aged Children survey,⁵⁴ asking how often the respondent had experienced pain, on average, during the last 6 months on a 5-point scale ranging from 0 ("rarely or never") to 4 ("about every day").

Pain interference was measured using 3 items developed for the purpose of the Three Cities Study, where respondents indicated whether during the last 6 months their pain severely interfered with 1) schoolwork, 2) leisure activities, and 3) peer relationships. The possible responses for each item were 2 ("no"), 1 ("yes, somewhat"), and 0 ("yes, definitely"). The 3 items were reversed and averaged into a total mean score ranging from 0 to 2.

Depressed Mood

The subscale "Depressed mood" of the Center for Epidemiology Studies' Depression Scale for Children was used to assess *depressed mood*. The full questionnaire consists of 20 items, each of which is rated from 0 ("not at all") to 3 ("a lot") and relates to symptoms perceived during the last week. At T1, depressed mood was mistakenly measured on a 1- to 5-point scale, which was rescaled to a 0- to 3-point scale to match the original scale, which was used from T2 to T5. The full scale and the subscales have been validated in Swedish adolescents,⁵⁵ where the same factor structure was identified in both girls and boys, and the cutoff for clinical depression is a total score of 24 for girls, and 22 for boys.⁵⁵ The subscale "Depressed mood" includes 6 items, such as "I felt down and unhappy." The internal consistency of the subscale in the current sample was $\alpha = .92$ for T1, $\alpha = .89$ for T2, $\alpha = .89$ for T3, $\alpha = .90$ for T4, and $\alpha = .90$ for T5.

Anxious Mood

The Overall Anxiety and Impairment Scale was used as a measure of *anxious mood*. Responses are made on a 5-item Likert scale, ranging from 0 ("none/no") to 4 ("constant/extreme/all the time"), concerning frequency of, intensity of, and functional impairment related to anxiety during the last week.⁵⁶ The cutoff score for clinical anxiety is 8.⁵⁷ The internal consistency in the study sample was $\alpha = .87$ for T1, $\alpha = .89$ for T2, $\alpha = .89$ for T3, $\alpha = .90$ for T4, and $\alpha = .91$ for T5.

Rumination

The depressive rumination subscale of the Children's Response Styles Scale⁵⁸ was used as a measure of rumination focusing on negative mood. The subscale contains 10 items that ask about rumination during the last 2 weeks, including "I think, 'why can't I stop feeling this way?'" and "I think about what made me feel like this." Responses range from 0 ("never") to 4 ("always"). The Children's Response Styles Scale has shown adequate validity in a sample of children, with an internal validity of $\alpha = .81$,⁵⁸ and has been used in Swedish adolescents.³⁸ The rumination item was included in the study survey from T1 to T4, but was excluded from the data collection at T5. In the current study sample, the internal consistency was $\alpha = .93$ for T1, $\alpha = .92$ for T2, $\alpha = .93$ for T3, and $\alpha = .93$ for T4.

Analysis Plan

Basic handling and preparation of data were performed in SPSS, version 28 (IBM Corp., Armonk, NY).⁵⁹ All of the subsequent analyses and structural equation modeling were performed in Mplus, version 8.9 (Muthén & Muthén, Los Angeles, CA),⁶⁰ and the power analysis was performed in R (R Core Team, Vienna, Austria).⁶¹ The statistical analysis was organized in 4 steps:

Step 1. Handling of Missing Data, Power Analysis, Descriptive Statistics, and Explanatory Analyses

To assess if we had sufficient power to conduct the analyses, we conducted the power analysis based on 2 complementary approaches: 1) based on general guidelines and rules-of-thumb, and 2) based on the MacCallum-Browne-Sugawara method. These assessments indicated that we have more than sufficient power to conduct the analyses, and they are detailed in [Supplementary Materials 1](#) (Section 1, "Power analysis"). Missing data have the potential to bias the data and results if there are systematic patterns in the missing data, which is imperative to address.⁶² In this study, missing data were handled by multiple imputation (MI) with refreshment/supplemental samples. MI is a state-of-the-art missing data handling method: it 1) creates datasets where the missing data are imputed on the basis of Bayesian probabilities, 2) fits the model/analysis to each imputed dataset, and 3) averages the estimates across the imputed datasets, and the estimates' standard errors are calculated according to Rubin's rules.⁶³ Refreshment or supplemental samples consist of study participants that did not participate at baseline but only on subsequent measurement occasions.^{64,65} Although these participants were not included in the analytic sample, their data were used in the missing data imputation process to reduce the potential bias of missingness. Including information from respondents of the same age and in the same context, who filled out the surveys at the same time as the baseline participants (analytic sample), enables more accurate estimations during imputation and thereby

reduces the potential negative impact of missing data.^{64,65} As can be seen in [Fig 1](#), the refreshment sample size was 440 at T2, 1,040 at T3, 1,592 at T4, and 1,501 at T5. Based on Enders' guidelines,⁶² we imputed 50 datasets. To avoid attenuating gender differences, we imputed the missing data on girls and boys separately.⁶⁶ As a sensitivity analysis, we also compared the estimated parameter values from MI, full information maximum likelihood estimation, and listwise deletion, described in [Supplementary Materials 1](#) (Section 2: "Sensitivity analysis of missing data handling methods" [Supplementary Table 1](#)).

After handling the missing data, we analyzed the sample characteristics of girls and boys separately. First, we examined the demographic and psychological variables at baseline. Second, we assessed the means and SDs of all of the main variables across all 5 measurement occasions. To assess differences across time and gender in these variables, we constructed linear latent growth curve models for each variable and compared differences in growth factors between the genders using Wald χ^2 -tests. Growth curve models assess mean differences at specific time points across the measured periods (intercept) and change in the variable over time (slope).⁶⁷ To assess mean differences at each time point, we moved the intercept from T1 through T5, and compared differences in the intercept factors at each time point in Wald χ^2 -tests. To assess differences in longitudinal change, we compared the slope factors across genders in Wald χ^2 -tests. The results from these analyses are detailed in [Supplementary Materials 1](#) (Section 3: "Gender-specific growth curve modeling on key variables" [Supplementary Table 2](#)).

The items in the measures of insomnia—depressed mood, anxious mood, and rumination—were parceled into 3 indicators per measure using a balancing approach where the item with the lowest item-scale correlation was paired with the item with the highest item-scale correlation; the item with the second-lowest correlation paired with the second-highest correlation; and so on. This is a recommended approach to approximate the optimal solution of item parcels.⁶⁶ The benefits of parceling include higher reliability, lower likelihood of distributional violations, and lower likelihood of correlated residuals.⁶⁶ These indicators were subsequently built into latent constructs with 3 parceled items per construct. A latent construct of pain was also constructed from the measures of pain intensity, pain frequency, and pain interference. Longitudinal measurement models were constructed for each latent variable, and strong measurement invariance was established for all models. We then tested competing models with the main constructs in order to establish which model fitted the data best while maintaining parsimony. Longitudinal full-mediation models were fitted with autoregressive paths and cross-sectional correlations between the constructs. We first explored various models with depressed mood, anxious mood, and rumination as mediators, and established an optimal model based on significant indirect effects and overall model fit. In this final model, insomnia was modeled as the predictor, rumination and depressed mood as serial mediators, and

pain as the outcome variable. These preliminary analyses are detailed in [Supplementary Materials 1](#) (Section 4: "Preliminary model testing," [Supplementary Table 3](#); [Supplementary Fig 1](#)). We then tested whether the model differed between girls and boys via multiple-group analysis using nested models. The model fit significantly worsened when keeping the model equal across gender. We, therefore, subsequently specified the model for girls and boys separately.

Step 2. Analysis of Measurement Invariance in the Final Model

Longitudinal confirmatory factor analyses were performed for all of the main constructs and then tested for both longitudinal measurement invariance and measurement invariance across gender. Specifically, we tested 3 types of measurement invariance: 1) *configural measurement invariance*, meaning that the model form was equal across time and gender; 2) *weak or metric invariance*, meaning that the factor loadings on the latent constructs were equal across time and gender; and 3) *strong or scalar invariance*, where the equality of indicator intercepts is tested across time and gender. If the assumption of both weak and strong measurement invariance holds, this procedure indicates whether the measures are addressing the same constructs across time and gender, and the differences and changes over time may be examined. Model goodness-of-fit was assessed on the basis of established guidelines^{66,68–70}: 1) model χ^2 with degrees of freedom and *P* values,⁷⁰ and 2) the Comparative Fit Index (CFI), which is an incremental fit index with values ranging from 0 to 1, where higher values represent a better fit (a CFI of at least .95 represents a good fit to the data⁶⁶). Also, 2 absolute fit indexes were used: 3) the root mean square error of approximation (RMSEA) with its 90% confidence interval, and 4) the standardized root mean square residual (SRMR). The absolute fit indices range in value from 0 to 1, where lower values represent a better fit.⁷⁰ A RMSEA of < .05 represents a good fit to data, as does a SRMR of .08 or less.⁶⁹ When assessing measurement invariance, a change in CFI of .01 or less, a change in RMSEA of .015 or less, and a change in SRMR of .03 or less were considered acceptable.⁶⁸ Given the large sample size, we expected the χ^2 -tests to be significant.⁶⁹ When testing nested models, χ^2 difference tests were utilized, where a significant test indicates that the nested model fits significantly worse than the less-restricted model.⁶⁶ Age, immigrant status, and SES were included as covariates in the analyses. As immigrant status and SES had nonsignificant associations with all of the main constructs, they were excluded from the final analyses.

Step 3: Longitudinal Mediation Modeling

Based on the findings of the previous steps, we fitted a longitudinal mediation model with insomnia as the predictor at T1, T2, and T3; rumination and depressed mood as serial mediators at T2, T3, and T4; and pain as the outcome at T3, T4, and T5. The model was

fitted for girls and boys separately. In line with the previous literature,^{66,71} which recommends avoiding analyzing specific mediation paths in favor of the more robust and reliable option of using the products of several mediation paths, we calculated the overall total, direct, and indirect effects (summing together all possible indirect effects from insomnia at T1 to pain at T5). The direct effect comprised the effect of insomnia on pain, excluding the mediation effects, and the indirect effects comprised the effects of 1) rumination, 2) depressed mood, and 3) both rumination and depressed mood. When calculating the total effects, all possible paths from insomnia to pain were summed. This procedure is described in greater detail in [Supplementary Materials 1](#) (Section 5: "Calculating the overall indirect effects" [Supplementary Fig 2](#)). The effect size of each indirect effect was calculated by dividing the indirect effect by the total effect, which gives the percentage of the total effect that the indirect effect accounts for. This method is appropriate for samples larger than 500.⁷² In the current study, we used the multivariate delta method to estimate the indirect effects, which is appropriate when using large samples⁷³ in which the distribution approaches asymptotic normality.⁷⁴ We also provide bias-corrected intervals, based on 10,000 bootstrap draws, in [Supplementary Materials 1](#) (Section 6: "Bootstrap confidence intervals of overall indirect effects" [Supplementary Table 4](#)).

Step 4. Analyses of Multigroup Moderation by Gender

Last, we tested the moderating effect of gender via multiple-group comparisons based on the longitudinal mediation model constructed in step 3, in which gender was dichotomized into girls and boys based on the baseline variable. First, we assessed differences in the overall effects via Wald χ^2 -tests. Then, we compared 6 nested models, each consisting of the effect of one construct on another (eg, insomnia on rumination), and assessed whether the nested model fitted significantly worse than the more unconstrained model. A significant difference in these analyses indicates moderation, that the examined effect is moderated by gender.⁷⁵

Results

Step 1: Sample Characteristics and Preliminary Analyses

As can be seen in [Table 1](#), the proportions of the sample of low SES and of an immigrant background were comparable across the genders. At baseline, there were significant but small age differences between girls and boys (χ^2 [5] = 13.22, *P* = .021). Compared with boys, considerably more girls scored above the clinical cutoff for insomnia. About half of the girls slept for a shorter time than recommended during weekdays, compared with a third of the boys. Girls also had higher rates of clinical depression (23.5%) and anxiety (18.1%) than

Table 1. Descriptive Information of the Baseline Sample at T1 ($M^{\text{age}} = 13.65$), Divided by Gender ($n^{\text{girls}} = 1,316$; $m^{\text{age girls}} = 13.63$, $n^{\text{boys}} = 1,451$; $m^{\text{age boys}} = 13.67$)

VARIABLE	GENDER		χ^2 (DF)	P
	GIRLS (%)	BOYS (%)		
Low SES	12.2	10.4	3.02 (1)	.082
Immigrant status			3.85 (3)	.278
Native Swedish	66.9	67.0		
Within EU	2.4	3.6		
Europe outside EU	7.5	6.6		
Outside Europe	23.2	22.8		
Clinical insomnia	26.0	16.5	37.68 (1)	< .001
Clinical depression	23.5	7.5	211.11 (1)	< .001
Clinical anxiety	18.1	5.5	107.23 (1)	< .001
Insufficient sleep duration weekdays (< 8 h)	48.3	31.6	93.27 (1)	< .001
Insufficient sleep duration weekends (< 8 h)	18.8	15.6	6.87 (1)	.009

Abbreviation: EU, European Union.

NOTE. The percentages refer to the proportions of boys and girls, separately. Immigrant status is categorized in accordance to the official Swedish definition. Differences between girls and boys are compared via χ^2 tests.

boys (clinical depression: 7.5%, clinical anxiety: 5.5%), in a magnitude of 3 to 4 times higher.

Table 2 details descriptive statistics for the key variables from T1 to T5, which are further detailed in [Supplementary Materials 1](#) (Section 6: "Gender-specific growth curve modeling on key variables") and the trajectories for insomnia and pain are visually represented in [Fig 2](#). Girls reported significantly increasing symptom

levels across the measured time period on all variables, and also reported significantly higher symptom levels than boys on all variables and at all time points. Boys reported significantly increasing symptom levels of insomnia, depressed mood, anxious mood, rumination, and pain frequency over time, but their pain interference and pain intensity did not significantly increase across the measured time period. Girls reported a significantly steeper rate of increase than boys on all the variables except for insomnia, where the rate of increase was not significantly different between the genders.

The preliminary analyses, which are detailed in [Supplementary Materials 1](#) (Section 3: "Preliminary model testing," and [Supplementary Table 2](#)), showed that insomnia predicted rumination and depressed mood, which, in turn, predicted pain. We also found that rumination was a stronger predictor of depressed mood than vice-versa ($\Delta\chi^2 [0] = 115.54$, $P < .001$), which confirmed our a priori hypothesis based on the response styles theory. Anxious mood was not a significant mediator of the effect of insomnia on pain, in neither girls nor boys, when either rumination or depressed mood was included in the model; it was only significant when included as a singular mediator in the model. Anxious mood was therefore excluded from the subsequent analyses. We then constructed a serial mediation model in which insomnia predicted subsequent rumination, which predicted depressed mood, which, in turn, predicted future pain. Estimating the path coefficients freely between girls and boys gave a significantly better data fit than that provided by a model that grouped girls and boys together ($\Delta\chi^2 [33] = 720.75$, $P < .001$). We, therefore, estimated separate models for girls and boys throughout the subsequent analyses.

Table 2. Descriptive Statistics for Key Variables at T1 to T5 by Gender

MEASURE/MEASUREMENT OCCASION (T)		T1	T2	T3	T4	T5
		M (SD)	M (SD)	M (SD)	M (SD)	M (SD)
Insomnia (0–28)*	Girls [†]	6.53 [‡] (5.35)	7.19 [‡] (5.61)	8.10 [‡] (5.94)	8.78 [‡] (6.22)	8.91 [‡] (6.44)
	Boys [†]	4.79 (2.19)	4.93 (2.22)	5.74 (2.39)	6.41 (2.53)	6.62 (2.57)
Pain frequency (0–4) [§]	Girls [†]	1.10 [‡] (1.26)	1.40 [‡] (1.38)	1.52 [‡] (1.41)	1.62 [‡] (1.44)	1.65 [‡] (1.46)
	Boys [†]	.87 (.93)	.94 (.97)	1.04 (1.02)	.99 (1.00)	.98 (.99)
Pain interference (0–2) [§]	Girls [†]	.27 [‡] (.43)	.32 [‡] (.47)	.32 [‡] (.48)	.34 [‡] (.49)	.36 [‡] (.53)
	Boys [†]	.24 (.48)	.22 (.46)	.23 (.48)	.21 (.46)	.21 (.46)
Pain intensity (0–9) [§]	Girls [†]	2.09 [‡] (2.53)	2.72 [‡] (2.69)	2.94 [‡] (2.85)	3.09 [‡] (2.81)	2.88 [‡] (2.81)
	Boys [†]	1.58 (1.26)	1.82 (1.35)	1.94 (1.39)	1.91 (1.38)	1.69 (1.30)
Depressed mood (0–18) [§]	Girls [†]	4.99 [‡] (4.51)	5.69 [‡] (4.75)	5.90 [‡] (4.60)	6.67 [‡] (4.90)	7.19 [‡] (4.84)
	Boys [†]	1.88 (1.37)	2.12 (1.46)	2.41 (1.55)	2.84 (1.69)	3.08 (1.76)
Anxious mood (0–20) [§]	Girls [†]	4.19 [‡] (3.90)	4.93 [‡] (4.35)	5.54 [‡] (4.58)	6.07 [‡] (4.82)	5.98 [‡] (4.89)
	Boys [†]	2.04 (1.43)	2.22 (1.49)	2.69 (1.64)	2.70 (1.64)	2.78 (1.67)
Rumination (0–40) [§]	Girls [†]	18.40 [‡] (8.69)	22.37 [‡] (8.55)	23.58 [‡] (8.88)	24.11 [‡] (8.33)	-
	Boys [†]	11.50 (3.39)	14.64 (3.83)	15.50 (3.94)	16.02 (4.00)	-

Abbreviation: M, mean.

NOTE. The values represent the estimated means after multiple imputation. Insomnia, depressed mood, anxious mood, and pain were measured at T1 through to T5. Rumination was measured at T1 to T4 only. Differences across time and gender were compared in latent growth curve models, which included all measures from T1 to T5 on all measures except for rumination, where the latent growth curve model was based on T1 to T4. Differences between the intercept growth factors were compared using Wald χ^2 tests, on all available time points. In all analyses, the cutoff for significance was set at $P < .05$.

[†]The differences were nonsignificant.

[‡]The longitudinal change in the variable was significant across the measured period.

[§]The mean level at this specific time point was significantly higher for this group than the other.

[§]The longitudinal rate of change was significantly higher in girls than in boys.

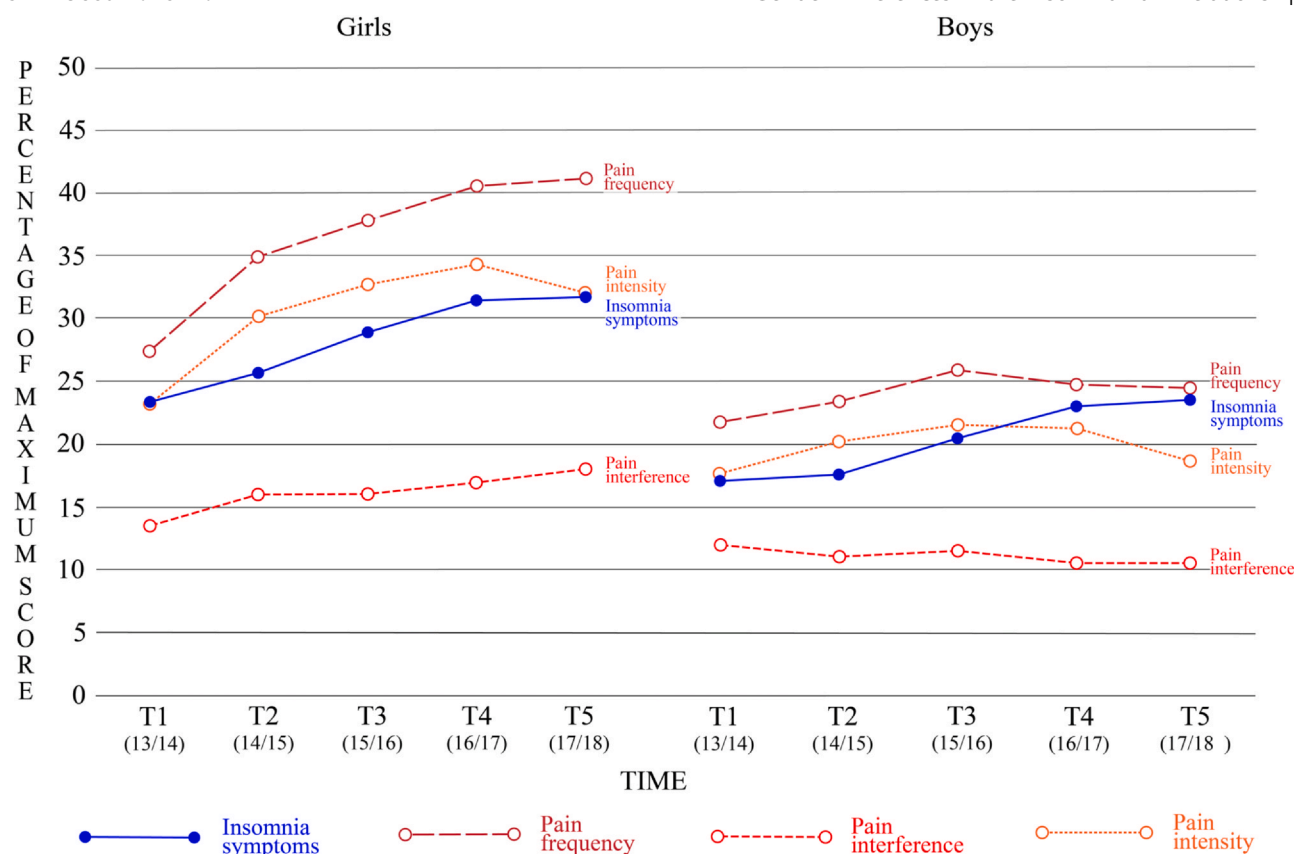


Figure 2. Depiction of changes in symptom levels of insomnia and pain variables from T1 to T5, in girls and boys separately. To facilitate comparison between variables and groups, the mean values are transformed to percentage of maximum score and presented on an axis from 0 to 50%.

Step 2: Analysis of Measurement Invariance in the Final Model

The longitudinal measurement model, including insomnia, rumination, depressed mood, and pain, met the assumption of strong measurement invariance, both across time and across the genders. The analyses of measurement invariance are detailed in [Table 3](#) below.

Step 3: Longitudinal Mediation Model

The final mediation model with insomnia as the predictor; rumination and depressed mood as serial mediators; and pain as the outcome, with the path coefficients freely estimated across gender and age included as covariates, showed a good fit to data ($\chi^2 [3,033] = 7,438.27$, $P < .001$; RMSEA = .028; CFI = .94; SRMR = .13). The high value of the SRMR was due to the age covariate.

The total effect of insomnia on pain across the measured period was significant both in girls ($b_{tot} = .41$, $P < .001$, 95% CI .313–.497) and in boys ($b_{tot} = .12$, $P < .001$, 95% (confidence interval) CI .054–.190). The direct effect of insomnia on pain was also significant in both girls ($b_{dir} = .33$, $P < .001$, 95% CI .217–.434) and boys ($b_{dir} = .11$, $P < .001$, 95% CI .034–.175).

The mediating (overall indirect) effect of depressed mood on the path from insomnia to pain was

nonsignificant in girls ($b_{ind} = .02$, $P = .126$, 95% CI –.006 to .052) but significant in boys ($b_{ind} = .02$, $P = .035$, 95% CI .001–.038). Depressed mood accounted for 6.9% of the total effect of insomnia on pain in girls and 16.4% in boys.

The mediating effect of rumination was significant and accounted for 14.1% of the total effect in girls ($b_{ind} = .06$, $P = .013$, 95% CI .012–.1029), but was not significant in boys ($b_{ind} > .01$, $P = .646$, 95% CI –.013 to .008), where it accounted for less than .01% of the total effect.

The serial mediation effect through both rumination and depressed mood was 6.4% of the total effect and nonsignificant in girls ($b_{ind} = .03$, $P = .086$, 95% CI –.004 to .055), but accounted for 9.0% of the total effect and was significant in boys ($b_{ind} = .01$, $P = .024$, 95% CI .001–.021).

Based on the statistical models that are detailed in [Fig. 3 and 4](#) below, it appears that insomnia consistently had an effect on both rumination and depressed mood; all of the specific paths from insomnia to pain were significant in both girls and boys, except for the effect of insomnia at T1 on rumination at T2 in boys. The effects of rumination and depressed mood on pain appear to be limited, however. In girls, it was only the specific effect of rumination at T3 on pain at T4 that was significant, and in boys, only the effect of depressed mood at T4 on pain at T5. [Table 4](#) summarizes the overall effects from insomnia at T1 to pain at T5.

Table 3. Model Fit Statistics for the Tests of Measurement Invariance in the Measurement Model Including Insomnia, Rumination, Depressed Mood, and Pain

MODEL TESTED	MEASUREMENT MODEL ESTIMATES							
	χ^2 (DF)	$\Delta\chi^2$ (Δ DF)	RMSEA (90% CI)	Δ RMSEA	CFI	Δ CFI	SRMR	Δ SRMR
<i>Across time (longitudinal invariance)</i>								
Null model	83,459.86 (1,596)	—	—	—	—	—	—	—
Configural invariance	2,235.49 (1,271)	—	.017 (.015–.018)	—	.988	—	.022	—
Metric (weak) invariance	2,371.53 (1,301)	136.04 (30)*	.017 (.016–.018)	< .001	.987	.001	.023	.001
Scalar (strong) invariance	2,649.79 (1,331)	278.26 (30)*	.019 (.018–.020)	.002	.984	.003	.024	.001
<i>Across gender (multigroup invariance)</i>								
Null model	75,625.75 (3,192)	—	—	—	—	—	—	—
Configural invariance	3,675.42 (2,542)	—	.018 (.017–.019)	—	.984	—	.028	—
Metric (weak) invariance	3,855.57 (2,580)	180.15 (38)*	.019 (.018–.020)	.001	.982	.002	.030	.002
Scalar (strong) invariance	4,032.07 (2,618)	176.50 (38)*	.020 (.019–.021)	.001	.980	.002	.030	< .001

NOTE. Measurement invariance was tested across time and across gender separately.
*Significant at $P < .001$.

Step 4: Multigroup Moderation by Gender

As the overall longitudinal mediation model fitted better when estimated freely between girls and boys ($\Delta\chi^2 [33] = 720.75$, $P < .001$), overall moderation of the effects is indicated. We therefore tested whether overall indirect effects, as well as specific tests, were moderated by gender.

The total effect of insomnia on pain was significantly stronger in girls (Wald $\chi^2 [1] = 23.79$, $P < .001$), compared with boys, as was the direct effect of insomnia on pain (Wald $\chi^2 [1] = 11.14$, $P < .001$). The overall indirect effect through rumination was significantly stronger in

girls (Wald $\chi^2 [1] = 6.57$, $P = .010$), but there were no significant differences between genders neither in the overall indirect effect through depressed mood (Wald $\chi^2 [1] = .03$, $P = .865$) nor the effect through both rumination and depressed mood (Wald $\chi^2 [1] = .861$, $P = .353$).

The specific effect of insomnia on pain was significantly moderated by gender ($\Delta\chi^2 [3] = 12.00$, $P = .007$), as was the effect of insomnia on rumination ($\Delta\chi^2 [3] = 14.93$, $P = .002$) but not on depressed mood ($\Delta\chi^2 [3] = 4.00$, $P = .261$). The effect of rumination on depressed mood was significantly moderated by gender ($\Delta\chi^2 [3] = 49.23$, $P < .001$), as was the effect of rumination on pain ($\Delta\chi^2 [3] = 11.05$, $P = .001$). Finally, the effect of depressed mood on pain was not significantly

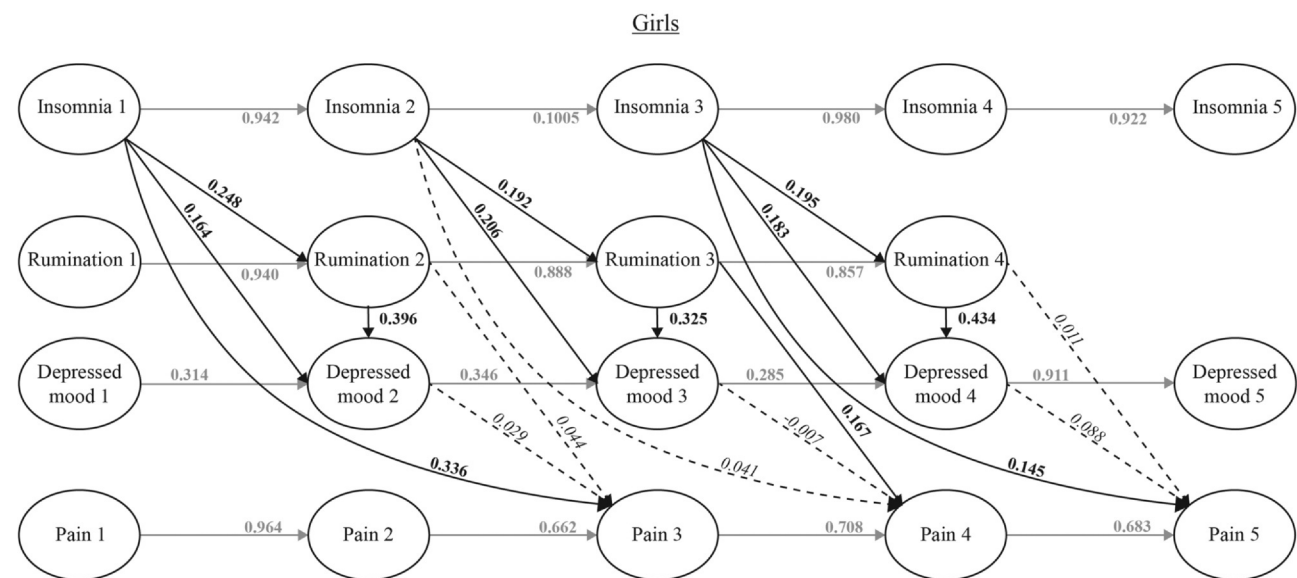


Figure 3. Depiction of the latent autoregressive longitudinal parallel mediation model in girls, with rumination and depressed mood as mediators and age as a covariate. To facilitate readability, cross-sectional correlations and covariate effects are omitted. Dotted lines represent nonsignificant paths.

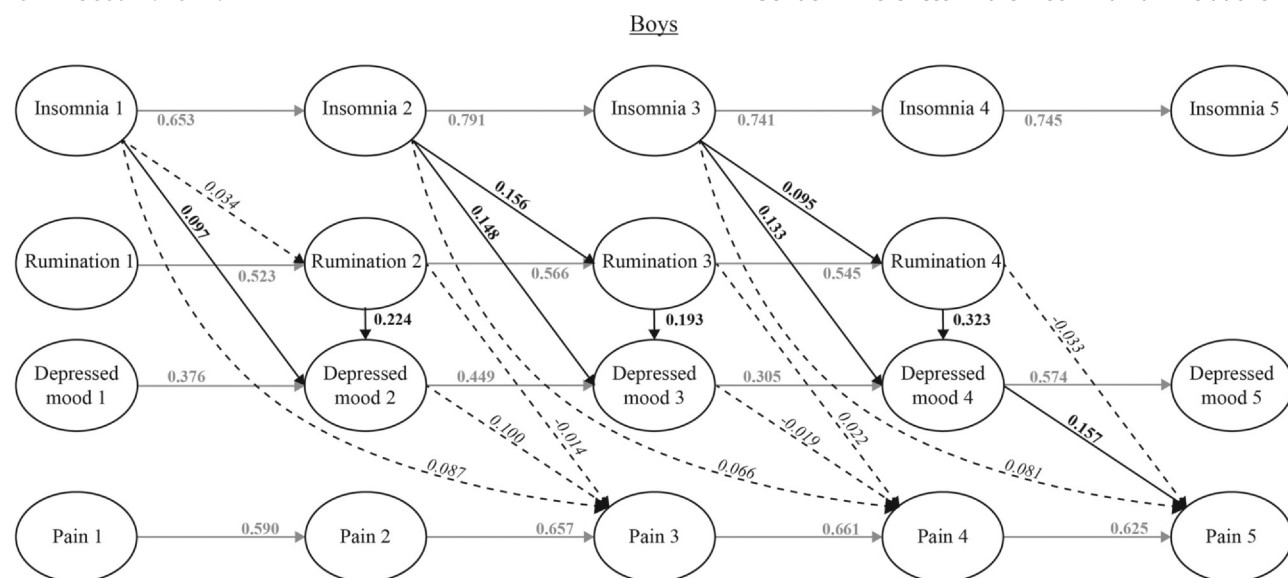


Figure 4. Depiction of the latent autoregressive longitudinal parallel mediation model in boys, with rumination and depressed mood as mediators and age as a covariate. To facilitate readability, cross-sectional correlations and covariate effects are omitted. Dotted lines represent nonsignificant paths.

Table 4. Overall Effects—Indirect, Direct, and Total—in Girls and Boys, as Well as Effect Sizes in Each of the Overall Effects

PATH	EFFECT			EFFECT SIZE
	B	P	95% CI	
<i>Total effect</i>				
Girls	.41*	< .001	.313 to .497	—
Boys	.12*	< .001	.054 to .190	—
<i>Insomnia-pain (direct)</i>				
Girls	.33*	< .001	.217 to .434	80.0
Boys	.11**	< .001	.034 to .175	86.0
<i>Insomnia-rumination-depressed mood-pain</i>				
Girls	.03 [†]	.092	-.004 to .055	6.4
Boys	.01**	.012	.001 to .021	9.0
<i>Insomnia-rumination-pain</i>				
Girls	.06**	.027	.012 to .102	14.1
Boys	> .01 [†]	.438	-.013 to .008	.0
<i>Insomnia-depressed mood-pain</i>				
Girls	.02 [†]	.102	-.006 to .052	6.9
Boys	.02**	.005	.001 to .038	16.4

NOTE. Coefficients in bold are significant. The summed effect sizes of the indirect and direct effects are larger than the total effects since the serial mediation of rumination and depressed mood overlaps with the single-mediator effects.

* $P < .001$.

** $P < .05$.

†Percentage of total effect.

‡Nonsignificant.

moderated by gender ($\Delta\chi^2 [3] = 1.27$, $P = .736$). Fig 5 offers a conceptual depiction of the effects that were moderated by gender.

Discussion

This study assessed how symptom levels of insomnia and pain changed during adolescence; how insomnia

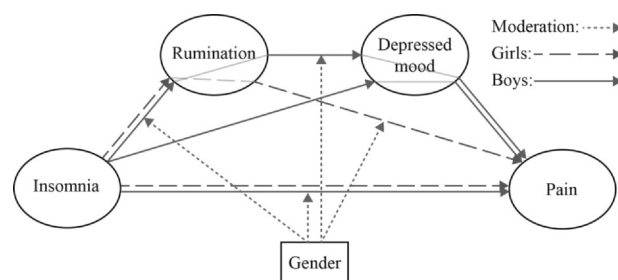


Figure 5. A simplified depiction of the moderated serial mediation model in the current study. The blue solid lines represent the significant overall effects in girls: 1) the direct effect from insomnia to pain, and 2) the indirect effect from insomnia, through rumination, to pain. The red dashed lines represent the significant overall effects in boys: 1) the direct effect from insomnia to pain, 2) the indirect effect from insomnia, through depressed mood, to pain, and 3) the indirect effect from insomnia, through both rumination and depressed mood, to pain. The green dotted lines represent the specific paths that were significantly moderated by gender. The effect of insomnia on rumination and pain, and also the effect of rumination on depressed mood and pain, were moderated by gender.

influenced pain during this time period; whether rumination and depressed mood could explain this influence; and whether this differed between boys and girls. Girls reported higher symptom levels of pain and insomnia compared with boys, although increases in insomnia were at a similar rate across the genders. Despite this, however, the negative effects of insomnia were particularly prominent in girls, as the total effect of insomnia on pain was 3.5 times stronger in girls, compared with boys. Based on the limited existing research, it appears that adolescence is an important developmental stage where connections between sleep and pain emerge^{14,25}; this is likely due to the biopsychosocial effects of puberty and adolescent development. After the onset of puberty, girls' pain seems to

become gradually more impacted by insomnia, a pattern continuing into adulthood.²⁷ At the same time, the effect of sleep disturbance on pain in boys seems to decrease after the onset of puberty.²⁸ Pubertal maturation is associated with both sleep disturbance and pain onset,^{76–78} and ample data have highlighted the impact of pubertal maturation on adolescent sleep, in particular, its disproportionate effect on females compared with males. For example, a meta-analysis demonstrated that females are 50% more likely to develop insomnia compared with males⁷⁹ and these stark differences are not evident prior to puberty, during when girls and boys have similarly low rates of insomnia. However, by mid-to-late puberty, girls have much higher rates of insomnia, and more severe symptoms than boys.^{80–82} There is evidence that sleep patterns change across the menstrual cycle. Adult females have reported poorer nighttime sleep quality, reduced sleep efficiency, shorter sleep duration, and daytime sleepiness during the premenstrual (late luteal) and menses (early follicular) phases compared with the late follicular and early luteal phases.⁸³ It is hypothesized that these changes are due to rapid reductions in progesterone and estrogen levels that exert influences on sleep quality, the circadian rhythm, and sleep architecture.^{83,84} These inherent hormonal fluctuations place females at higher risks of repeated and cyclical disruptions to their regular sleep patterns, and could help to explain the increased rate of sleep disturbances that begin after onset of puberty. Further research is needed to understand the additional causal mechanisms that explain the differential relationship between sleep and pain in boys and girls.

The finding that girls report higher symptom levels and increases compared with boys was not surprising, given that adolescent girls commonly report more insomnia and pain than boys.^{1,7} What was more surprising was how much stronger the effect of insomnia on pain was in girls, and how insomnia influenced pain differently in boys and girls. Contrary to our initial hypothesis, the response styles theory did not explain the stronger effect of insomnia on pain in adolescent girls, as only rumination, but not depressed mood, mediated the effect of insomnia on pain in girls. Rather the alternative hypothesis, that rumination directly mediates the relationship via its functional overlap with pain-catastrophizing, seems to be supported by our data: insomnia may increase rumination by affecting attentional control and emotional states,²⁴ which, in turn, leads to more pain and related suffering.^{85,86} Pain-catastrophizing is closely related to rumination,⁸⁷ and to both pain and sleep.^{88,89} It appears that the ruminative aspects of pain-catastrophizing are particularly salient in the sleep-pain relationship.⁸⁸ Furthermore, ruminative thinking is related to cognitive biases.^{90,91} A cognitive model of comorbid insomnia and pain emphasizes cognitive processes in the sleep-pain relationship.⁹² Although not a focus in the model, it may be

particularly relevant in females. In addition, cognitive presleep arousal overlaps with rumination^{93,94} and predicts greater pain^{43,95} independently of depressed mood.^{94,95} This distinction of sleep-related rumination may help to explain why depressed mood was not a mediator in girls, which is contrary to the findings of previous studies.^{15–17} Notably, no previous study has examined the effect of depressed mood in girls and boys separately; nor did they include rumination as a simultaneous mediator.

Depressed mood influenced the sleep-pain relationship in boys despite the fact that girls reported a 3 times higher prevalence of clinical depression, and that the effect of insomnia on depressed mood, and on depressed mood on pain, did not differ across genders. One hypothesis is that the dopaminergic system plays a particularly important role in males, as sleep deprivation, chronic pain, and depressed mood are all associated with the dopaminergic system.^{96,97} There are sex differences in dopaminergic functioning,⁹⁷ and males are overrepresented in psychiatric disorders where dopamine plays an important role.⁹⁸ Interestingly, sleep problems are a common comorbidity in the same disorders.⁹⁹ However, as the effects in boys were small and boys reported limited increases in pain, and the abovementioned research has been conducted primarily in adults, the conclusions that may be drawn from this finding are limited.

Given that anxious mood has been identified as a mediator in the sleep-pain path in previous studies,^{17,100,101} its nonsignificant effect was unexpected. The mediating effect of anxious mood may be better explained by overlapping aspects of depressed mood and rumination rather than the unique facets of anxious mood.

Despite rumination and depressed mood being significant mediators, the strongest effects remained the direct effects, explained by unmeasured mechanisms. These mechanisms may include neurophysiological systems,¹⁰² as females are more affected by sleep disruption in their endogenous pain modulation than males, and lowered pain inhibition due to sleep deprivation appears to only occur in females, not in males.¹⁰³ Increased inflammation, higher cortisol levels, and hypothalamic-pituitary-adrenal axis dysfunction, caused by sleep disturbance, are also associated with more severe pain,^{25,97,104} and these associations are stronger in females than males.^{105,106} Importantly, these processes appear to be especially prominent during adolescence, and particularly in adolescent girls.¹⁴

This study's results might have practical implications, particularly given that adolescence seems to be a critical developmental stage to prevent both insomnia and chronic pain.^{14,107} Although research on cognitive behavior therapy for insomnia in adolescents with comorbid chronic pain is scarce, emerging results are promising, showing reduced insomnia in adolescents with comorbid pain and other physical and psychiatric

conditions.^{108,109} Perhaps, such interventions may be enhanced by incorporating techniques targeted at rumination, which has been shown to be related to improvements in a broad range of outcomes in previous research.¹¹⁰ The gender differences we observed could mean that boys and girls each require unique intervention approaches to optimize treatment outcome, and may partly explain why insomnia treatments targeted at comorbid chronic pain have shown promising but inconsistent results.^{111,112} Increasing research on the sleep-pain relationship, which appears to reach beyond the walls of academia,¹¹³ will hopefully translate into more effective early interventions for comorbid sleep and pain problems in the future.

This study has several strengths. It is based on a large population sample spanning most of adolescence, from 13/14 to 17/18 years, with an equal proportion of girls and boys. It utilized latent variables, which controls for measurement error that may otherwise have biased the results.⁶⁶ We considered mediation paths at multiple time points, which is a robust method of examining longitudinal mediation,⁷¹ and also examined multiple mediators simultaneously, allowing us to partial out the cardinal mediator effects in overlapping constructs. Finally, we examined moderation via multiple-group comparisons, which allowed us to analyze gender differences in the insomnia-pain relationship to an extent and detail not previously reported.

A limitation of this study is that it includes only self-reported measures. This is particularly relevant regarding the binary gender variable, since it neither directly measures sex assigned at birth nor the complexities of gender identity. Future research with more precise measures of sex and gender is warranted. Objective sleep measures, such as actigraphy, are important to include in future studies, since subjective and objective sleep measures address overlapping but distinct aspects of sleep.¹¹⁴ A T5 measure of rumination would have been ideal, although largely extraneous, as the last mediator path at T4 had previous values controlled for via autoregressive effects. The varying timeframes of the included measures are also suboptimal. However, due to the yearly measurements and type of analysis used in the study,^{66,71} in combination with all constructs being trait-like,^{115–118} we believe the results mainly reflect stable, trait-like effects. Although a considerable proportion of girls self-reported as above the clinical cutoffs for insomnia (26%), depression (23.5%), and anxiety (18.1%), to facilitate further generalization of these results, we recommend that future studies attempt to replicate the study findings in clinical populations. Beyond insomnia disorder, chronic pain, and mental disorders, this also includes additional clinical populations, such as persons suffering from cancer. A lack of information on race and ethnicity is also a limitation, as these factors have been shown to be associated with pain¹¹⁹ and insomnia¹²⁰ in

previous studies. Our closest proxy of these constructs is immigrant status, which was included.

In conclusion, we found that adolescent girls were more vulnerable to the effects of insomnia on pain compared with boys. Insomnia also influenced pain differently across the genders, with rumination having an important role in girls, and depressed mood in boys. These results have important implications for both theoretical and clinical research. Examining the insomnia-pain relationship in girls and boys separately is necessary to correctly understand how insomnia influences pain, and if treatments for comorbid insomnia and pain may benefit from being tailored to girls and boys separately.

Data availability

In [Supplementary Materials 2](#) (Section 1, “Mplus syntax”), we provide the Mplus syntax for our analyses. We also provide the variance-covariance matrix that our analyses were based upon ([Supplementary Materials 2](#), Section 2, “Variance-covariance matrix”), which allows for replication of the analyses. The raw data set may be accessed following permission from Katja Boersma (katja.boersma@oru.se), PI of the Three Cities Study.

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Disclosures

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Appendix A. Supplementary Data

Supplementary data related to this article can be found at [doi:10.1016/j.jpain.2024.104571](https://doi.org/10.1016/j.jpain.2024.104571).

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